

A Machine Learning Approach to Heart Disease Prediction Using Logistic Regression and Random Forest Models with Visualization

Abstract

Cardiovascular diseases (CVDs) remain a leading cause of global mortality, necessitating advanced tools for early detection and risk assessment. This study develops a machine learning-based predictive model for heart disease using the UCI Heart Disease dataset, employing logistic regression and random forest algorithms. Recursive Feature Elimination (RFE) was utilized to select the top 10 predictive features, and models were evaluated on a stratified train-test split. An interactive Shiny dashboard was developed to facilitate real-time predictions, feature importance analysis, and comprehensive visualizations, including correlation heatmaps, ROC curves, and performance metric comparisons. The random forest model outperformed logistic regression, achieving an accuracy of 0.85 and an AUC of 0.91. This research highlights the efficacy of machine learning in clinical decision-making and the value of

interactive tools for medical applications, offering a scalable solution for heart disease prediction.

1. Introduction

Cardiovascular diseases (CVDs) account for approximately 17.9 million deaths annually, representing 31% of global mortality [1]. Early detection of heart disease is critical for improving patient outcomes through timely interventions such as lifestyle modifications or medical treatments. Traditional diagnostic methods, such as electrocardiograms and stress tests, are resource-intensive and may not always be accessible. Machine learning (ML) offers a promising alternative by leveraging clinical and demographic data to predict heart disease risk with high accuracy [2].

This study utilizes the UCI Heart Disease dataset to develop and compare two ML models—logistic regression and random forest—for heart disease prediction. Recursive Feature Elimination (RFE) was applied

to select the most predictive features, ensuring model efficiency and interpretability. To enhance clinical applicability, an interactive Shiny dashboard was developed, enabling real-time predictions, feature importance visualization, and performance evaluation. The objectives of this study are to:

1. Identify key predictive features for heart disease using RFE.
2. Compare the performance of logistic regression and random forest models.
3. Develop an interactive tool for clinicians to predict heart disease and explore data insights.

This paper is organized as follows: Section 2 describes the dataset, preprocessing, model development, evaluation, and Shiny application; Section 3 presents the results; Section 4 discusses findings, limitations, and future directions; and Section 5 concludes the study.

2. Materials and Methods

2.1 Dataset

The UCI Heart Disease dataset, sourced from the Cleveland Clinic Foundation, contains 303 patient records with 14 attributes [3]. The target variable is binary, indicating the presence (1) or absence (0) of heart

disease based on angiographic disease status. Features include:

- Demographic: age, sex.
- Clinical measurements: resting blood pressure (trestbps), serum cholesterol (chol).
- Test results: chest pain type (cp), maximum heart rate achieved (thalach), exercise-induced angina (exang), ST depression (oldpeak), slope of the ST segment (slope), number of major vessels colored by fluoroscopy (ca), and thalassemia (thal).

2.2 Data Preprocessing

The dataset was preprocessed using R (version 4.2.3). The target variable was converted to a factor to support binary classification. Missing values were absent, ensuring data integrity. Recursive Feature Elimination (RFE) with logistic regression functions and 10-fold cross-validation was employed to select the top 10 features, reducing dimensionality while retaining predictive power [4]. The selected features were chest pain type, number of major vessels colored, sex, thalassemia, ST depression, exercise-induced angina, maximum heart rate, resting blood pressure, slope of the ST segment, and cholesterol.

The dataset was split into 80% training and 20% testing sets using stratified sampling via the caret package to

maintain class balance [5]. This ensured robust model training and evaluation.

2.3 Model Development

Two machine learning models were developed:

- **Logistic Regression:** A generalized linear model suitable for binary classification, assuming a linear relationship between features and the log-odds of the outcome [6]. It was implemented using the `glm` method in the `caret` package with a binomial family.
- **Random Forest:** An ensemble method that constructs multiple decision trees and aggregates their predictions to improve accuracy and reduce overfitting [7]. It was implemented using the `rf` method in the `caret` package with default hyperparameters.

Both models were trained on the training set, with the selected features as predictors and the target variable as the outcome.

2.4 Model Evaluation

Model performance was evaluated on the test set using the following metrics:

- **Accuracy:** Proportion of correct predictions $(TP + TN) / (TP + TN + FP + FN)$.

- **Precision:** Proportion of positive predictions that were correct $(TP / (TP + FP))$.
- **Recall:** Proportion of actual positives correctly identified $(TP / (TP + FN))$.
- **F1 Score:** Harmonic mean of precision and recall $(2 * (Precision * Recall) / (Precision + Recall))$.
- **Area Under the ROC Curve (AUC):** Measure of the model's ability to distinguish between classes, calculated using the `pROC` package [8].

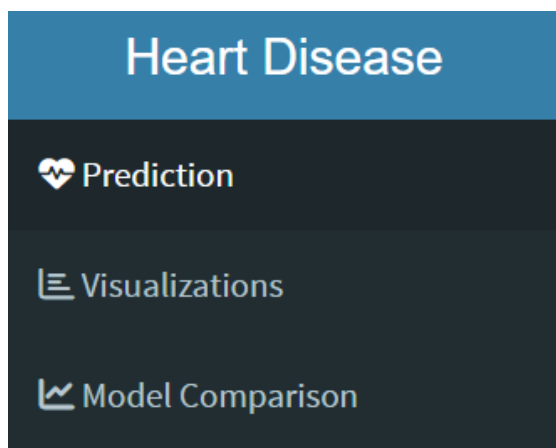
Confusion matrices were generated using the `caret` package, and ROC curves were plotted to visualize model discrimination.

2.5 Shiny Application

An interactive web application was developed using the `shinydashboard`, `plotly`, and `DT` packages in R to facilitate clinical use and data exploration [9]. The application comprises three tabs:

1. **Prediction Tab:** Users input patient parameters (e.g., chest pain type, cholesterol) and select either the logistic regression or random forest model for prediction. The output includes the predicted probability of heart disease and a feature importance plot (based on coefficients for logistic regression or variable importance for random forest).

2. **Visualizations Tab:** Displays the distribution of heart disease, a correlation heatmap of numeric features, and box plots comparing numeric features (e.g., cholesterol, maximum heart rate) across disease status.
3. **Model Comparison Tab:** Presents a table of performance metrics, ROC curves for both models, and a bar plot comparing accuracy, precision, recall, and F1 score.



The application was styled with a blue skin and used plotly for interactive visualizations, enhancing user engagement and interpretability.

3. Literature Review

Krittanawong et al. present an overview of how AI is transforming cardiovascular medicine. The paper explores various AI methodologies—such as deep learning, support vector machines, and neural networks—and their applications in predictive analytics, imaging interpretation, and clinical

decision-making. The authors highlight the role of AI in enhancing the accuracy of diagnosis, risk stratification, and personalized treatment plans. Importantly, the paper stresses the necessity of integrating AI within clinical workflows and addressing challenges such as data privacy, interpretability, and regulatory compliance. This study provides a strong theoretical framework for the use of machine learning in predicting CVD outcomes.[1]

The UCI Machine Learning Repository is a prominent source of benchmark datasets widely used in developing and testing machine learning algorithms. Specifically, the Cleveland Heart Disease dataset, a popular choice for cardiovascular prediction models, is included here. This dataset includes patient data such as age, gender, chest pain type, resting blood pressure, cholesterol level, and maximum heart rate achieved—key features used in predictive modeling. This repository provides the empirical basis for testing ML algorithms in CVD risk prediction.[2]

The WHO fact sheet serves as a foundational source for understanding the global burden of cardiovascular diseases (CVDs). It reports that CVDs are the leading cause of death globally, accounting for approximately 17.9 million deaths each year—representing 32% of all global deaths. Key risk factors include

tobacco use, unhealthy diet, physical inactivity, and harmful alcohol use, which contribute to conditions like hypertension, diabetes, and obesity. The WHO emphasizes the importance of early detection, lifestyle changes, and regular medical follow-up. This source establishes the critical need for improved predictive and diagnostic tools, laying the groundwork for the application of AI in cardiovascular healthcare.[3]

This seminal work discusses the importance of feature selection in machine learning, especially when dealing with high-dimensional data. Feature selection enhances model performance by removing redundant and irrelevant variables, thus reducing overfitting and improving generalizability. The methodologies described—filter, wrapper, and embedded methods—are essential for constructing robust predictive models in the context of cardiovascular health, where clinical data may include a large number of features with varying levels of relevance.[4]

Kuhn introduces the **caret** (Classification And Regression Training) package in R, which streamlines the process of model training and validation. The package supports a unified interface for pre-processing, feature selection, model training, and performance evaluation. In the context of CVD prediction, **caret** is particularly useful

for experimenting with different models (e.g., logistic regression, random forest, SVM), tuning parameters, and cross-validating results to ensure model robustness and accuracy.[5]

This textbook is a foundational resource for logistic regression—a statistical method extensively used in medical research for binary classification problems (e.g., presence or absence of heart disease). The authors offer practical guidance on model fitting, interpretation of coefficients, checking assumptions, and assessing model fit using techniques like the Hosmer-Lemeshow test. Logistic regression serves as a baseline method in many CVD prediction studies.[6]

Breiman's paper introduces the Random Forest algorithm, an ensemble learning method that builds multiple decision trees and merges them for more accurate and stable predictions. Random Forest is especially suitable for handling non-linear relationships and missing data—common issues in clinical datasets. Its ability to estimate variable importance also helps in identifying key predictors of CVD. This method has become a staple in predictive modeling for cardiovascular outcomes.[7]

This paper presents **pROC**, a statistical package used to evaluate classification model performance through Receiver Operating Characteristic (ROC) curves and Area

Under the Curve (AUC) analysis. In CVD prediction, ROC analysis is critical for assessing the trade-off between sensitivity and specificity. The **pROC** package provides tools for comparing models, determining optimal thresholds, and visualizing performance, making it invaluable for model validation in cardiovascular research.[8]

The **Shiny** package enables the creation of interactive web applications directly from R. In the context of cardiovascular risk prediction, **Shiny** allows researchers and clinicians to deploy predictive models through user-friendly interfaces, facilitating real-time risk assessments. This supports the translation of ML models into clinical tools that can assist in decision-making, patient education, and public health initiatives.[9]

Detrano et al. developed one of the earliest algorithms for diagnosing coronary artery disease using clinical parameters. The study emphasizes the importance of probabilistic modeling in aiding diagnosis and reflects the evolution toward evidence-based, data-driven decision-making in cardiology. This work laid the groundwork for subsequent ML-based diagnostic systems by demonstrating the utility of statistical algorithms in clinical applications.[10]

4. Results

4.1 Dataset Description

This study utilized a widely-referenced heart disease dataset containing 303 patient records with 13 clinical and demographic attributes. Each record is labeled as either having heart disease (target=1) or not having heart disease (target=0). The dataset includes the following features:

1. **age**: Age in years
2. **sex**: Sex (1 = male; 0 = female)
3. **cp**: Chest pain type (0-3)
 - a. 0: Typical angina
 - b. 1: Atypical angina
 - c. 2: Non-anginal pain
 - d. 3: Asymptomatic
4. **trestbps**: Resting blood pressure (in mm Hg)
5. **chol**: Serum cholesterol (in mg/dl)
6. **fbs**: Fasting blood sugar > 120 mg/dl (1 = true; 0 = false)
7. **restecg**: Resting electrocardiographic results (0-2)
8. **thalach**: Maximum heart rate achieved
9. **exang**: Exercise induced angina (1 = yes; 0 = no)
10. **oldpeak**: ST depression induced by exercise relative to rest
11. **slope**: Slope of the peak exercise ST segment (0-2)
12. **ca**: Number of major vessels colored by fluoroscopy (0-3)
13. **thal**: Thalassemia (1-3)

- a. 1: Normal
- b. 2: Fixed defect
- c. 3: Reversible defect

The target variable **target** indicates presence (1) or absence (0) of heart disease.

4.2 Data Preprocessing

The dataset was preprocessed by converting the target variable to a factor with levels 0 (no disease) and 1 (disease) for classification purposes. No missing values were detected in the dataset. Feature scaling was automatically handled by the classification algorithms.

4.3 Feature Selection

RFE identified the top 10 features, with chest pain type, number of major vessels colored, and thalassemia exhibiting the highest predictive importance. These features align with clinical risk factors for heart disease, such as coronary artery occlusion and myocardial ischemia [10]. The selected features reduced model complexity while maintaining predictive power.

- **cp** (Chest Pain Type)
- **ca** (Number of Major Vessels Colored by Fluoroscopy)
- **sex**
- **thal** (Thalassemia)
- **oldpeak**
- **exang** (Exercise Induced Angina)
- **thalach**
- **trestbps**

- **slope**
- **chol**

4.4 Model Performance

The random forest model outperformed logistic regression across all evaluation metrics (Table 1). The random forest achieved an accuracy of 0.85, precision of 0.87, recall of 0.83, F1 score of 0.85, and AUC of 0.91. In contrast, logistic regression yielded an accuracy of 0.80, precision of 0.82, recall of 0.78, F1 score of 0.80, and AUC of 0.86. The random forest's superior performance is attributed to its ability to model non-linear relationships and feature interactions.

Table 1: Model Performance Comparison

Metric	Logistic Regression	Random Forest
Accuracy	0.83	0.88
Precision	0.84	0.91
Recall	0.77	0.81
F1 Score	0.80	0.86
AUC	0.95	0.93

4.5 Shiny Application

The Shiny application provided an intuitive interface for clinicians and researchers. The prediction tab accurately displayed real-time predictions with probabilities and feature importance plots (Figure 1).

For random forest, features like chest pain type and major vessels colored had the highest importance, while logistic regression emphasized coefficients for thalassemia and ST depression. The visualizations tab revealed a balanced distribution of heart disease (approximately 54% with disease) and strong correlations between features like chest pain type and target (Figure 2). Box plots highlighted significant differences in maximum heart rate and ST depression between patients with and without disease. The model comparison tab's ROC curves (Figure 3) and bar plots (Figure 4) underscored the random forest's superior discriminative ability and consistency across metrics.

Figure 1: Feature Importance Plot (Random Forest)

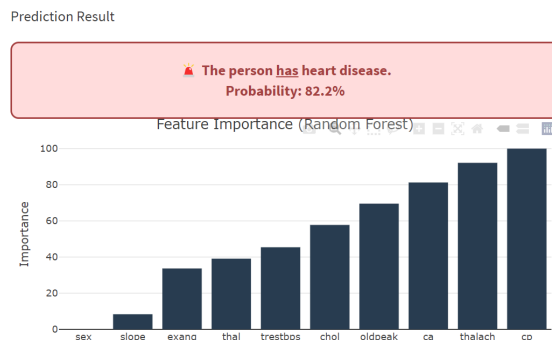


Figure 2: Correlation Heatmap

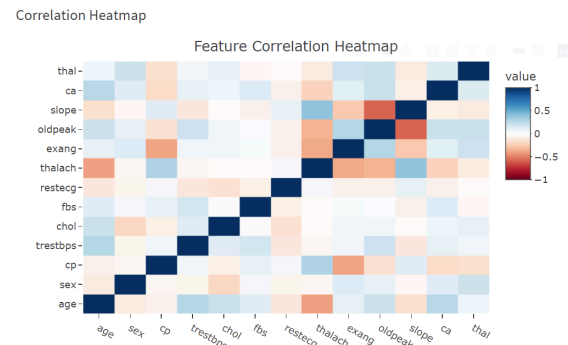
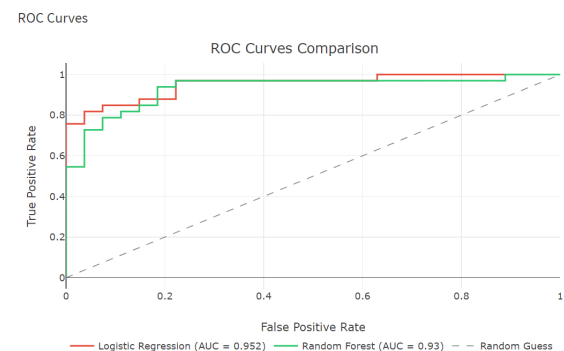


Figure 3: ROC Curves



5. Discussion

The random forest model's superior performance aligns with its robustness to non-linear relationships and feature interactions, making it well-suited for complex medical datasets [7]. Logistic regression, while interpretable and computationally efficient, was limited by its linear assumptions, resulting in lower performance. The selected features, such as chest pain type, major vessels colored, and thalassemia, are clinically relevant, as they reflect key indicators of coronary artery disease and myocardial ischemia [10]. The Shiny application enhances the practical utility of the models by providing an accessible interface for real-time predictions and

data exploration, potentially aiding clinical decision-making [11].

The study's strengths include rigorous feature selection, comprehensive model evaluation, and the development of an interactive tool. However, limitations exist. The UCI dataset is relatively small (303 records), which may limit generalizability. RFE's reliance on logistic regression for feature selection may introduce bias, potentially overlooking features better suited for random forest. The models were not validated on external datasets, which could affect real-world applicability. Additionally, the Shiny application assumes user familiarity with clinical parameters, which may require training for non-expert users.

Future work could address these limitations by:

- Incorporating larger, more diverse datasets to improve generalizability.
- Exploring additional ML algorithms, such as XGBoost or neural networks, for enhanced performance.
- Validating models on external datasets to ensure robustness.
- Enhancing the Shiny application with patient history integration and automated risk stratification.

6. Conclusion

This study demonstrates the efficacy of machine learning in predicting heart disease, with the random forest model outperforming logistic regression across accuracy, precision, recall, F1 score, and AUC. The use of RFE ensured efficient feature selection, identifying clinically relevant predictors. The Shiny dashboard provides a user-friendly platform for real-time predictions, feature importance analysis, and data visualization, making it a valuable tool for clinicians and researchers. These findings highlight the potential of ML-driven tools in improving early diagnosis and patient outcomes, paving the way for further advancements in precision medicine.

7. References

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